

Education

New York University

PH.D STUDENT IN ELECTRICAL AND COMPUTER ENGINEERING

- Advisor: Prof. Austin Rovinski
- Lab: REALISE Lab

New York, USA

Aug 2026 - Present

Seoul National University of Science and Technology

M.S IN ELECTRONIC ENGINEERING

- Advisor: Prof. Seung Eun Lee
- Lab: Computer Architecture Laboratory
- Thesis: Approximate Arithmetic Circuits for Embedded Fuzzy Logic Controller

Seoul, Republic of Korea

Mar 2022 - Feb 2024

Seoul National University of Science and Technology

B.S IN ELECTRONIC ENGINEERING

- Advisor: Prof. Seung Eun Lee

Seoul, Republic of Korea

Mar 2015 - Feb 2022

Work Experience

MangoBoost

SOFTWARE ENGINEER

- Developed a Kubernetes-native storage orchestration platform (Go Operator, Helm, React dashboard) managing DAOS distributed storage on bare-metal clusters.
- Implemented node lifecycle automation including NVMe/SPDK device management, SCM memory sizing, and multi-engine hugepage allocation.
- Operated and deployed multi-node K8s clusters with end-to-end provisioning pipeline (kubespray, Docker, Helm).

Seoul, South Korea

Jan 2026 - July 2026

SYSTEM ARCHITECT

- Architected an NVMe/TCP initiator (NTI) with dynamic ARM-FPGA acceleration, allowing protocol processing to be flexibly offloaded depending on performance requirements and hardware availability. This ensured both efficiency and reliability in large-scale deployments (e.g., Ceph).
- Extended scalability from the original 2 subsystems to 18 FPGA-accelerated functions via SR-IOV, while achieving full 200G line-rate throughput in aggregated performance tests. On the ARM side, designed a flexible execution path supporting up to 130 concurrent functions to enable diverse application workloads.
- Verified functionality and throughput using FIO benchmarks across multiple configurations (single disk, RAID, XFS).
- Measured real-world database performance with PostgreSQL (pgbench) workloads.

Mar 2024 - Dec 2025

Selected Publications [Full list on Google Scholar]

- **BoostX™-NTI: Fast, Scalable and Flexible Storage Architecture with NVMe/TCP Initiator Acceleration** [URL]
The 53rd International Symposium on Computer Architecture (ISCA) Industry Track, 2026
Hamin Jang, Jinha Jeong, Wonseok Lee, Jun Heo, Jongcheon Lee, Hyunjae Chu, Taehyun Kim, **Youngwoo Jeong**, Dongu Kim, Heetaek Jeong, Changsu Kim, Dongup Kwon, Jangwoo Kim
- **SEAM: A synergetic energy-efficient approximate multiplier for application demanding substantial computational resources** [URL]
Integration. vol.101, 2025
Youngwoo Jeong, Jounghmin Park, Raehyeong Kim, Seung Eun Lee
- **Robot-Specific Processor for Autonomous Driving**
1st Workshop on Robotics Acceleration with Computing Hardware (RoboARCH) (Co-located with MICRO), Chicago, USA, Oct., 2022
Youngwoo Jeong, Kwang Hyun Go, Soohee Kim, Seung Eun Lee
- **An Architecture for Resilient Federated Learning through Parallel Recognition (Poster Session)** [URL]
The 31st International Conference on Parallel Architectures and Compilation Techniques (PACT), Chicago, USA, Oct., 2022
Jeongeun Kim, **Youngwoo Jeong**, Suyeon Jang, Seung Eun Lee
- **Photoplethysmography-Based Distance Estimation for True Wireless Stereo** [URL]
Micromachines. vol.14, no.2, 2023
Youngwoo Jeong, Jounghmin Park, Sun Beom Kwon, Seung Eun Lee

Patents

Federated Learning Method and System Using Shared Learning Data

SEUNG EUN LEE, JEONGEUN KIM, **YOUNGWO JEONG**

patent application

United States

December 2023

Method and System for Determining Final Result Using Federated Learning

SEUNG EUN LEE, JEONGEUN KIM, **YOUNGWO JEONG**

patent application

United States

December 2023

Research Project

Development for Processing Software on AI Semiconductor Devices MINISTRY OF SCIENCE AND ICT	<i>South Korea</i> 2024 - 2022
- Analyzed various AI models to standardize the input for AI systems. - Proposed an architecture for a hardware scheduler optimized for multi-AI core architecture.	
Development of Proximity/Healthcare Convergence Sensor SoC for TWS MINISTRY OF TRADE, INDUSTRY AND ENERGY	<i>South Korea</i> 2023 - 2021
- Designed a test environment for photoplethysmography sensors to evaluate their performance. - Proposed a waveform adjustment filter and AI-based distance estimation algorithm to enhance signal processing accuracy.	
Embedded AI Module Based on Neuromorphic Computing MINISTRY OF TRADE, INDUSTRY AND ENERGY	<i>South Korea</i> 2021 - 2020
- Designed various applications utilizing multiple embedded AI modules. - Developed a testbed for evaluating multi-AI core controllers. - Proposed methodologies to enhance accuracy in federated learning with multi-AI core systems.	

Honors & Awards

School of Engineering Ph.D. Fellowship NEW YORK UNIVERSITY (NYU)	<i>New York, USA</i> August 2026
Excellent Thesis Award SEOUL NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY	<i>Seoul, South Korea</i> February 2024
Topic: Approximate Arithmetic Circuits for Embedded Fuzzy Logic Controller	
Corporate (LX Semicon) Special Award KOREA SEMICONDUCTOR INDUSTRY ASSOCIATION	<i>Seoul, South Korea</i> October 2022
Topic: AI Processor employing Stochastic Computing for Embedded Systems	
Department Chair Award SEOUL NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY	<i>Seoul, South Korea</i> February 2022
Topic: Design of an Autonomous Indoor Robot for Robot-on-Chip	
Corporate (Silicon Mitus) Special Award KOREA SEMICONDUCTOR INDUSTRY ASSOCIATION	<i>Seoul, South Korea</i> November 2021
Topic: In-Vehicle Network Processor based on LIN and CAN-FD Controller	

Teaching Experience

Advanced AI Processor, Computer Architecture SEOUL NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY Teaching Assistant	<i>Seoul, South Korea</i> Fall 2022
Digital System Design, Resilient Processor Design SEOUL NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY Teaching Assistant	<i>Seoul, South Korea</i> Spring 2022

Skills

Hardware Description Languages	Verilog, SystemVerilog
High-Level Computer Languages	SystemC, C, C++, Python, Matlab
Design and Implementation Tools	Catapult HLS, Design Compiler, IC Compiler II, Quartus II, Vivado
Verification and Analysis Tools	Verdi, VCS, ModelSim, PSpice, PrimeTime, Formality, StarRC
Benchmark Tools	Flexible I/O (FIO), PGbench

Chip Design

Design of Robot-Specific Processor for Autonomous Driving
- Technology: Samsung 28nm RFCMOS - Designer: Youngwoo Jeong, Yue Ri Jeong, Hyun Woo Oh, Kwang Hyun Go - Gate Counts: 1062K @ 50MHz - Memory: Code region (16KB), Data region (128KB) - Date: 2022. 07. 18

